

Department of Computer Science and Engineering
Indian Institute of Technology, Hyderabad



**Executive MTech in Data Science
Admission Test – Objective Exam**

Date: 20-06-2016
Time: 10:30–12:30 pm

TOTAL MARKS: 80

Please fill-in the following details:

Name _____

Application ID _____

PLEASE READ BEFORE ANSWERING:

1. The first shortlisting would be done based on this objective exam. Only for candidates who score more than a minimum threshold mark in this exam, we would grade the Background in Data Science exam that would be held in the afternoon.
2. This paper has two problems: Problem 1 and Problem 2.
3. There are 30 MCQs in Problem 1. For each MCQ choose the right answer from (a), (b), (c), (d) and mark it on the answer sheet on page 6. Each of these questions has exactly one correct answer. Each correct answer carries a mark of **+2** and wrong answer carries a mark of **-0.5**.
4. Problem 2 requires you to fill in a missing value by seeing the number pattern. Each correct answer carries a mark of **+5**. There is no negative marking.
5. During the exam you are allowed to use a SAT-approved calculator.

[60 Points] Problem 1: Multiple Choice Questions

Questions 1-30 has exactly **ONE** correct answer. (Correct answer: **+2**, Wrong answer: **-0.5**)

1. For the following JAVA code, which of the following is correct:

```
public void foo(boolean a, boolean b)
{
    if(a)
        System.out.println("A");
    else if(a && b)
        System.out.println("A && B");
    else {
        if(!b)
            System.out.println("notB") ;
        else
            System.out.println("ELSE") ;
    }
}
```

- a. If a is true and b is true then the output is "A && B"
- b. If a is true and b is false then the output is "notB"
- c. If a is false and b is true then the output is "ELSE"
- d. If a is false and b is false then the output is "ELSE"

2. What will be the output of the program (in Java)?

```
try {  
    int x = 0;  
    int y = 5 / x;  
}  
catch (Exception e){  
    System.out.println("Exception");  
}  
catch (ArithmeticException ae){  
    System.out.println("Arithmetic Exception");  
}  
System.out.println("Finished");
```

- a. Finished
 - b. Compilation fails
 - c. Exception
 - d. Arithmetic Exception
3. Which of the following is not the member of class (in C++)?
- a. Static function
 - b. Const function
 - c. Friend function
 - d. Virtual function
4. Which of the following statement is correct?
- a. A constructor is called at the time of declaration of an object.
 - b. A constructor is called at the time of use of an object.
 - c. A constructor is called at the time of declaration of a class.
 - d. A constructor is called at the time of use of a class.
5. Which of the following functions are performed by a constructor?
- a. Construct a new class
 - b. Construct a new function
 - c. Construct a new object
 - d. Initialize objects
6. Which of the following statement is correct (in C++)?
- a. Overloaded functions can have at most one default argument.
 - b. An overloaded function cannot have default argument.
 - c. All arguments of an overloaded function can be default.
 - d. A function if overloaded more than once cannot have default argument.
7. Three unbiased coins are tossed. What is the probability of getting at most two heads?
- a. 3/4
 - b. 3/8
 - c. 1/4
 - d. 7/8
8. In a class, there are 15 boys and 10 girls. Three students are selected at random. The probability that 1 girl and 2 boys are selected, is:
- a. 21/46
 - b. 1/50
 - c. 25/117
 - d. 3/25
9. Suppose that you have to read a sequence of length n of 0s and 1s. You must keep track of the number of ones that you have seen. How many bits of memory do you need?
- a. $\Theta(\log n)$
 - b. $\Theta(n^2)$
 - c. $\Theta(n)$
 - d. $\Theta(2^n)$
10. Consider a binary search tree with n vertices and height h . Which of the following is the asymptotic worst case time complexity for inserting a new element into this tree?
- a. $O(\log h)$
 - b. $O(h)$
 - c. $O(\log n)$
 - d. $O(n)$
11. If $k \neq 0$, then $\gcd(ka, kb)$ is
- a. $\gcd(a, b)$
 - b. $|k| \gcd(a, b)$
 - c. $k \gcd(a, b)$
 - d. 1

12. Which of the following statement is correct?
- Overloaded functions can accept same number of arguments.
 - Overloaded functions always return value of same data type.
 - Overloaded functions can accept only same number and same type of arguments.
 - Overloaded functions can accept only different number and different type of arguments.
13. Algorithm A has Algorithm B as its subroutine. The running time of algorithm A without counting the time spent running the algorithm B subroutine is $O(n^4)$. One call of algorithm B takes $O(n^3)$ time. In a single run of algorithm A, algorithm B is called at most $10n^2$ times. What is the running time of algorithm A including subroutines?
- $O(n^4)$
 - $O(n^9)$
 - $O(n^7)$
 - $O(n^6)$
14. The rank of the matrix $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$ is:
- 1
 - 3
 - 2
 - 9
15. Which of the following is **FALSE** about K-Means clustering:
- It creates hierarchical clusters
 - It creates exclusive clusters (i.e., each data point belongs to a single cluster)
 - It can be used for data with numerical attributes
 - It uses Euclidean similarity measure
16. Which of the following is **NOT** used to connect a program with a DBMS server:
- JDBC
 - SQL
 - ODBC
 - ADO.NET
17. Suppose relation $R(A,B,C)$ currently has only the tuple $(0,0,0)$, and it must always satisfy the functional dependencies $A \rightarrow B$ and $B \rightarrow C$. Which of the following tuples may be inserted into R legally?
- $(0,0,2)$
 - $(1,0,1)$
 - $(0,2,0)$
 - $(1,1,0)$
18. Consider the following query on the relation $R(A, B, C, D)$
- ```
SELECT []
FROM R
GROUP BY A, B content...
```
- Assume  $A, B, C, D$  take integer values. Which of the following can appear in the position marked as [...]:
- $\text{MIN}(C+D)$
  - $A, B$
  - $C, D$
- II only
  - I, II, and III
  - I and II only
  - None
19. Consider two (strong) entity sets and a many-to-many relationship between them in an E-R diagram. When mapped to the relational model, this design will result in:
- Exactly 2 tables
  - Exactly 3 tables
  - At most 2 tables
  - A single table
20. Consider a relation  $R(A,B)$  with  $r$  tuples, all unique within  $R$ , and a relation  $S(B,C)$  with  $s$  tuples, all unique within  $S$ . Let  $t$  represent the number of tuples in  $R$  natural-join  $S$ . Which of the following triples of values  $(r,s,t)$  is possible?
- $(2, 3, 8)$
  - $(2, 3, 9)$
  - $(3, 3, 10)$
  - $(3, 3, 0)$

21. What is the maximum size (in number of tuples) of the result of expression  $R \text{ join } S$ , if the join attribute is a candidate key of relation  $S$ ?

- a.  $n_R$  (number of tuples in  $R$ )
- b.  $\min(n_S, n_R)$
- c.  $n_S$
- d.  $\max(n_S, n_R)$

22. The following SQL statement over tables  $R(a,b)$ ,  $S(b,c)$ , and  $T(a,c)$  requires certain privileges to execute:

```
UPDATE R
SET a = 10
WHERE b IN (SELECT c FROM S)
AND NOT EXISTS (SELECT a FROM T WHERE T.a = R.a)
```

Which of the following privileges is not useful for execution of this SQL statement?

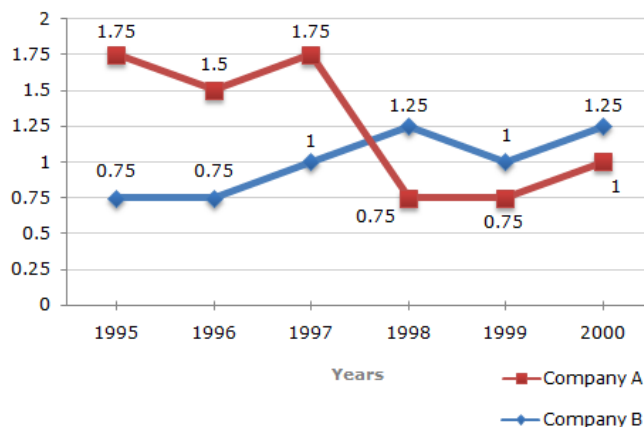
- a. SELECT ON  $R$
- b. UPDATE ON  $R(a)$
- c. UPDATE ON  $S$
- d. SELECT ON  $T$

23. Suppose you have a table Employee with two columns: ID and Salary. What does the following SQL query compute? Assume that in the table there is no repetition in the ID values, and no repetition in the Salary values as well.

```
SELECT ID
FROM Employee E1
WHERE (2) = (
SELECT COUNT(E2.Salary)
FROM Employee E2
WHERE E2.Salary > E1.Salary)
```

- a. Finds the ID of the employee having the third highest salary.
- b. Finds the ID of the employee having the highest salary.
- c. Finds the ID of the employee having the lowest salary.
- d. Finds the ID of the employee having the second lowest salary.

**Answer Questions 24-28 based on the following line graph.**



**Figure 1:** Ratio of Exports to Imports (in terms of money in Rs. crores) of Two Companies over the Years

24. In how many of the given years were the exports more than the imports for Company A?

- a. 2
- b. 4
- c. 3
- d. 5

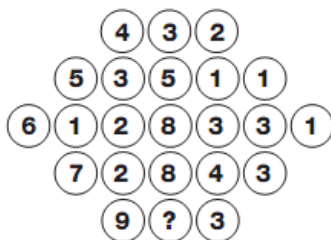
25. If the imports of Company A in 1997 were increased by 40 percent, what would be the ratio of exports to the increased imports?
- 1.20
  - 1.30
  - 1.25
  - cannot be determined
26. If the exports of Company A in 1998 were Rs. 237 crores, what was the amount of imports in that year?
- Rs. 189.6 crores
  - Rs. 281 crores
  - Rs. 243 crores
  - Rs. 316 crores
27. In 1995, the export of Company A was double that of Company B. If the imports of Company A during the year was Rs. 180 crores, what was the approximate amount of imports of Company B during that year?
- Rs. 190 crores
  - Rs. 225 crores
  - Rs. 210 crores
  - Cannot be determined
28. In which year(s) was the difference between imports and exports of Company B the maximum?
- 2000
  - 1998 and 2000
  - 1996
  - Cannot be determined

**For Questions 29 and 30, find the statement that must be true according to the given information.**

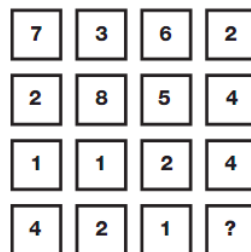
29. Erin is twelve years old. For three years, she has been asking her parents for a dog. Her parents have told her that they believe a dog would not be happy in an apartment, but they have given her permission to have a bird. Erin has not yet decided what kind of bird she would like to have.
- Erin's parents like birds better than they like dogs.
  - Erin does not like birds.
  - Erin and her parents live in an apartment.
  - Erin and her parents would like to move.
30. Ten new television shows appeared during the month of September. Five of the shows were sitcoms, three were hour-long dramas, and two were news-magazine shows. By January, only seven of these new shows were still on the air. Five of the shows that remained were sitcoms.
- Only one of the news-magazine shows remained on the air.
  - Only one of the hour-long dramas remained on the air.
  - At least one of the shows that was cancelled was an hour-long drama.
  - Television viewers prefer sitcoms over hour-long dramas.

**[20 points] Question 2: Number Patterns [Correct answer: +5]**

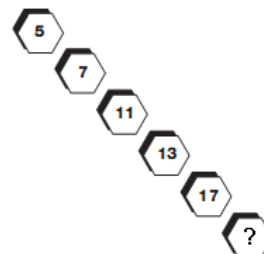
For each of the following, which number replaces the question mark?



? =



? =



? =



? =

## Problem 1 Answers

Please enter a tick mark  $\checkmark$  for the correct choice for each question. Each question has exactly **one** correct choice.

| Question Number | (a) | (b) | (c) | (d) |
|-----------------|-----|-----|-----|-----|
| 1               |     |     |     |     |
| 2               |     |     |     |     |
| 3               |     |     |     |     |
| 4               |     |     |     |     |
| 5               |     |     |     |     |
| 6               |     |     |     |     |
| 7               |     |     |     |     |
| 8               |     |     |     |     |
| 9               |     |     |     |     |
| 10              |     |     |     |     |
| 11              |     |     |     |     |
| 12              |     |     |     |     |
| 13              |     |     |     |     |
| 14              |     |     |     |     |
| 15              |     |     |     |     |
| 16              |     |     |     |     |
| 17              |     |     |     |     |
| 18              |     |     |     |     |
| 19              |     |     |     |     |
| 20              |     |     |     |     |
| 21              |     |     |     |     |
| 22              |     |     |     |     |
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| 24              |     |     |     |     |
| 25              |     |     |     |     |
| 26              |     |     |     |     |
| 27              |     |     |     |     |
| 28              |     |     |     |     |
| 29              |     |     |     |     |
| 30              |     |     |     |     |